

GOVATHOTI VAMSI

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Professional Summary

Software-focused Electronics and Communication Engineering graduate with hands-on experience in Python, Java, SQL, JavaScript, and AI/ML application development. Strong foundation in object-oriented programming, data structures, DBMS, debugging, exception handling, and software engineering fundamentals. Built practical software, web, and AI applications using Python, TensorFlow, OpenCV, Streamlit, React.js, RAG, LLMs, and backend technologies, with experience developing real-time computer vision solutions.

Experience

AI/ML Intern – APSCHE & SmartInternz

April 2025 – July 2025 — Remote

- Developed a CNN-based fruit and vegetable classification system using Python, TensorFlow, and OpenCV on a dataset containing 32,000+ images across 26 classes, achieving 97% classification accuracy.
- Performed image preprocessing, data augmentation, model training, validation, and evaluation while debugging class-label mapping and prediction inconsistencies in the classification pipeline.
- Tested and refined classification outputs across multiple fresh and rotten fruit and vegetable categories to validate prediction performance and improve inference reliability.
- Repository:** github.com/vamsigovathoti8125/smart-sorting-transfer-learning-for-identifying-rotten-fruits-and-vegetables

Education

Seshadri Rao Gudlavaluru Engineering College

2022 – 2026

B.Tech in Electronics and Communication Engineering

CGPA: 7.69/10

Projects

vamsiMart – E-Commerce Web Application

- Developed a multipage e-commerce website using HTML5, CSS3, JavaScript, Node.js, Express.js, and SQLite with product browsing across multiple categories.
- Implemented product search, category filtering, price sorting, load-more pagination, product details, persistent shopping carts, and responsive layouts.
- Built registration and login functionality with JWT authentication, account-specific carts, checkout with multiple payment options, order creation, and order history.
- Repository:** github.com/vamsigovathoti8125/ecommerce-vamsiMart

StudyMind AI – Agentic RAG Study Assistant

- Built a Python and Streamlit application that processes uploaded PDF documents and provides context-aware answers using retrieved document content.
- Engineered the document processing pipeline using PDF processing, text chunking, embeddings, FAISS vector search, and semantic retrieval.
- Achieved 94% query-answering accuracy with response times below 15 seconds using query classification, agent-based routing, and LLM-powered responses.
- Repository:** github.com/vamsigovathoti8125/studymindai-rag

Job Application Tracker – Frontend Web Application

- Built a responsive web application using HTML5, CSS3, and JavaScript to organize and track job applications across recruitment stages.
- Created features for adding, updating, viewing, and organizing application details using DOM manipulation, event handling, and dynamic UI updates.
- Applied form validation, frontend debugging, and responsive design principles to improve application workflows and usability.
- Repository:** github.com/vamsigovathoti8125/job-application-tracker

AI-Based Driver Drowsiness and Distraction Detection System

- Built a real-time driver monitoring application using Python, OpenCV, Dlib, ESP32, and Blynk.
- Applied facial landmark detection with Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to identify eye closure and yawning.
- Achieved 92% detection accuracy while integrating fatigue, distraction, and alcohol-sensing alerts with ESP32 and mobile notifications.

Technical Skills

Programming: Python, Java, C, JavaScript

Core Programming: OOP, Data Structures, Algorithms, Problem Solving, Debugging, Exception Handling

Backend Development: Flask, FastAPI, REST API Fundamentals, API Development, Deployment

Database: SQL, MySQL, DBMS, Database Fundamentals

Web Technologies: HTML5, CSS3, JavaScript, React.js, Tailwind CSS

AI/ML: Machine Learning, Deep Learning, CNN, Computer Vision, TensorFlow, OpenCV, Dlib

Generative AI: RAG, LLMs, Agentic AI, Prompt Engineering, Embeddings, Semantic Search, FAISS

Frameworks & Tools: Streamlit, Git, GitHub, Git Branching, VS Code

Computer Science: Operating Systems, Computer Networks, Software Development, Model Evaluation

Cloud & Systems: AWS Fundamentals, CI/CD Fundamentals, Linux Fundamentals

My Portfolio: vamsigovathoti8125-portfolio-app-s2t5h6.streamlit.app